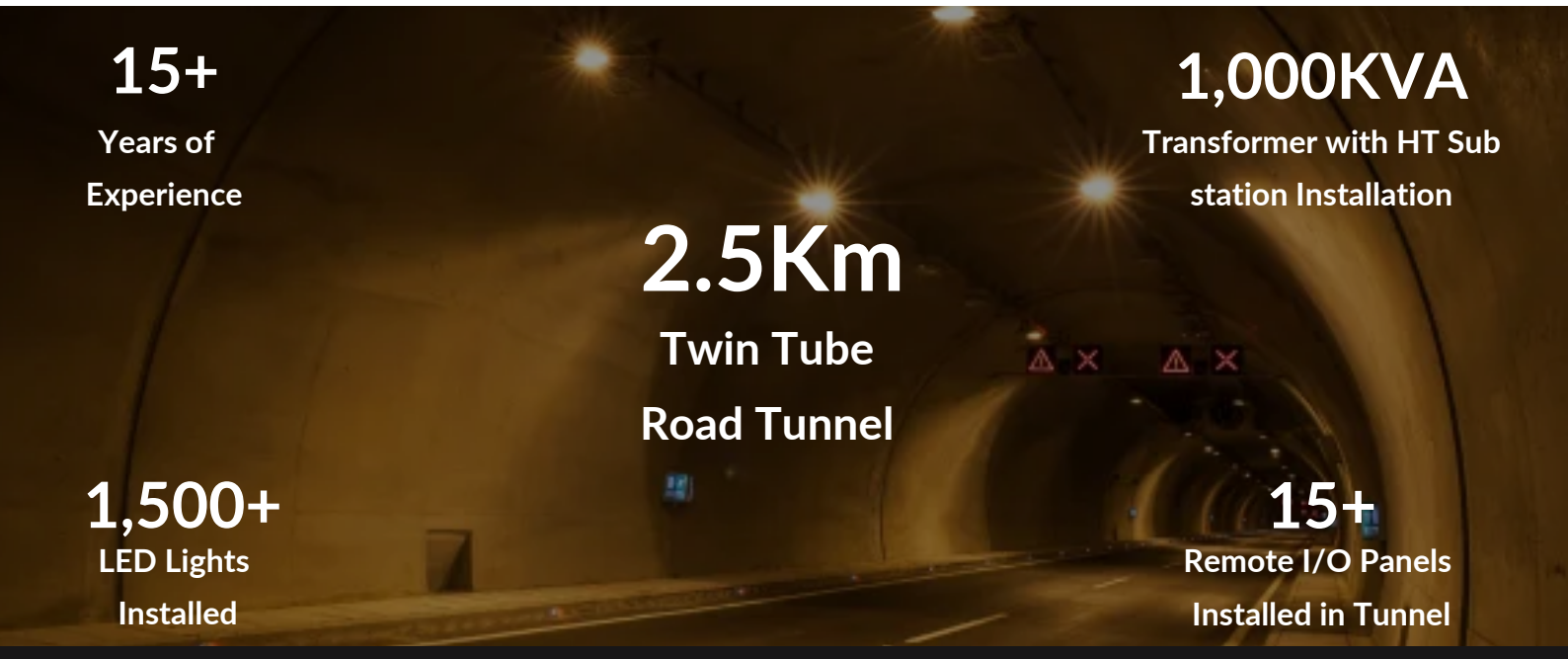
Adaptive Engineering Pvt. Ltd.
RESULT ORIENTED ▪ SERVICE DRIVEN

Tunnel Solutions

Smart Tunnels for a Smarter Future



15+
Years of Experience

1,000KVA
Transformer with HT Sub station Installation

2.5Km
Twin Tube Road Tunnel

1,500+
LED Lights Installed

15+
Remote I/O Panels Installed in Tunnel

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Tunnel Solutions



- Tunnel control centre (SCADA)
- Power distribution system
- Tunnel lighting
- Tunnel ventilation
- Fire control
- CCTV
- Video Automatic Incident Detection (VAID)
- Dynamic messaging and signalling
- Public Address Voice Alarm System (PAVA)

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SCADA

SCADA Monitors & Controls systems are used in tunnel management to monitor and control various aspects such as:

- Ventilation systems
- Lighting systems
- Security and surveillance
- Fire detection and suppression
- Traffic management
- Environmental monitoring





SCADA helps in:

- Monitor real-time data and alarms
- Control operations remotely
- Optimize operations and energy consumption
- Improve safety and security
- Respond to emergencies quickly

Power Distribution System

- Reliable power supply
- Redundant systems for backup
- Distribution panels and transformers
- Cabling and busbars
- Power monitoring and control systems
- Backup power sources
- Surge protection devices
- Power quality monitoring



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Tunnel Lighting

Normal Operations

- Enhanced safety and security
- Improved visibility and navigation
- Emergency response and evacuation
- Energy efficiency
- Cost savings
- Smoother traffic flow
- Reduced congestion
- Reduced carbon footprint



Emergency Operations

- Lighting through power backup in case of power outage
- Evacuation lighting in case of emergency

Tunnel Ventilation

Normal Operations

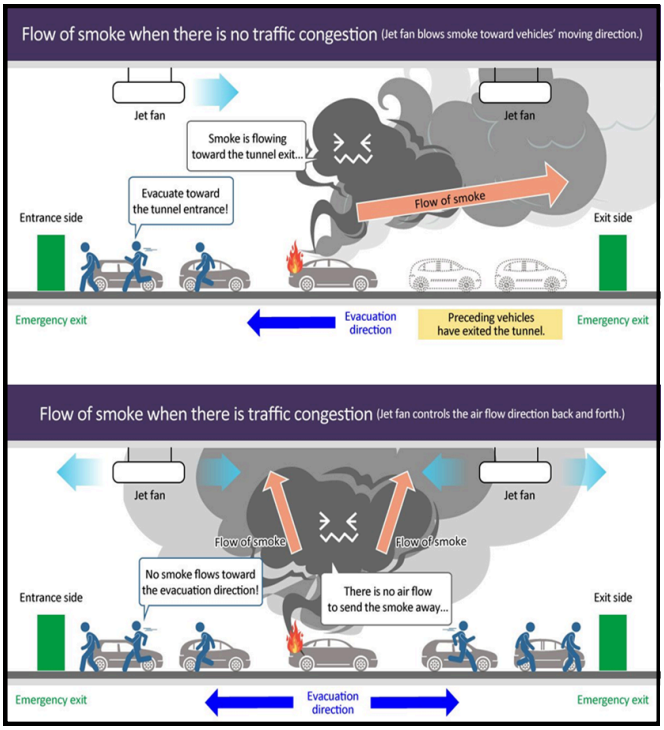
- Improved air quality
- Enhanced safety
- Improved emergency response
- Energy efficiency
- Cost savings
- Extended lifespan of tunnels
- Removal of harmful gases and particles
- Improved driver visibility and comfort
- Protection from fire



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Tunnel Ventilation

- Advanced detection systems (smoke, heat, flame)
- Automated suppression systems (water, foam, gas) manual override and activation
- Integration with SCADA and tunnel management systems
- Real-time monitoring and alerting
- Fire pump and water supply management
- Emergency lighting and communication systems
- Regular testing and maintenance scheduling
- Compliance with industry standards and regulations



Fire Control



Safety functions for fire detection

- Advanced detection systems (smoke, heat, flame)
- Automated suppression systems (water, foam, gas) manual override and activation
- Integration with SCADA and tunnel management systems
- Real-time monitoring and alerting
- Fire pump and water supply management
- Emergency lighting and communication systems
- Regular testing and maintenance scheduling
- Compliance with industry standards and regulations

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CCTV

- Real-time monitoring
- High-resolution video quality
- Night vision and low-light sensitivity
- Motion detection and alerts
- Automatic number plate recognition (ANPR)
- Video analytics (e.g., object detection, crowd monitoring)
- Deterrent to criminal activity
- Improved traffic flow and management
- Real-time monitoring for emergency services



Dynamic Messaging & Signaling





Providing road tunnel user information about:

- Lane allocation
- Speed limits
- Access barriers
- Emergency information
- Evacuation route leading to emergency exits
- Public awareness
- Tunnel closure Information
- Alternative route guidence

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Video Automatic Incident Detection (VAID)

Several algorithms are available from the VAID system which can detect a range of incidents, including:

- Stopped vehicles
- Smoke
- Vehicles moving in the wrong direction
- Speed reduction
- Slow vehicle
- Pedestrians
- Debris in the road tunnel
- Flames
- Entry into restricted zones
- Lost goods in the tunnel



Figure 1: Congestion Detection in a two-lane tunnel



Figure 7: Pedestrian on road

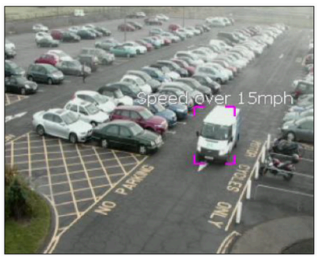
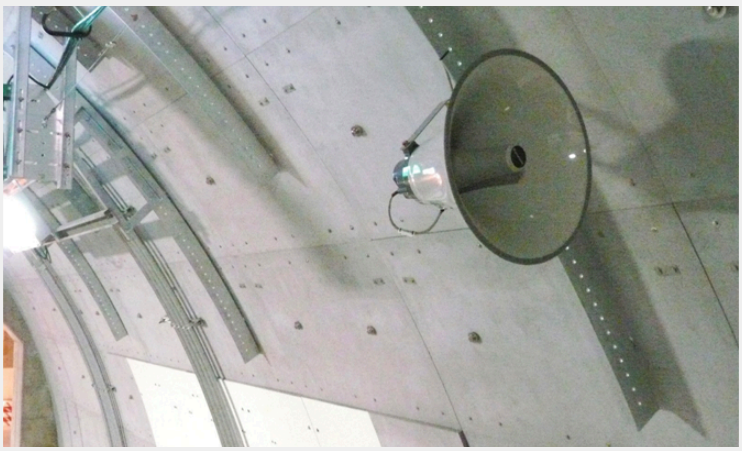


Figure 4: Speeding Detection



Figure 9: Ipsotek's Visual Smoke Detection in a tunne

Public Address Voice Alaram System (PAVA)



- Emergency communication and alerts
- Safety instructions and guidance
- Traffic management and updates
- Emergency services coordination and response
- Public information and announcements
- Evacuation instructions and guidance
- Fire alarm notifications and alerts
- Tunnel closure information and notifications

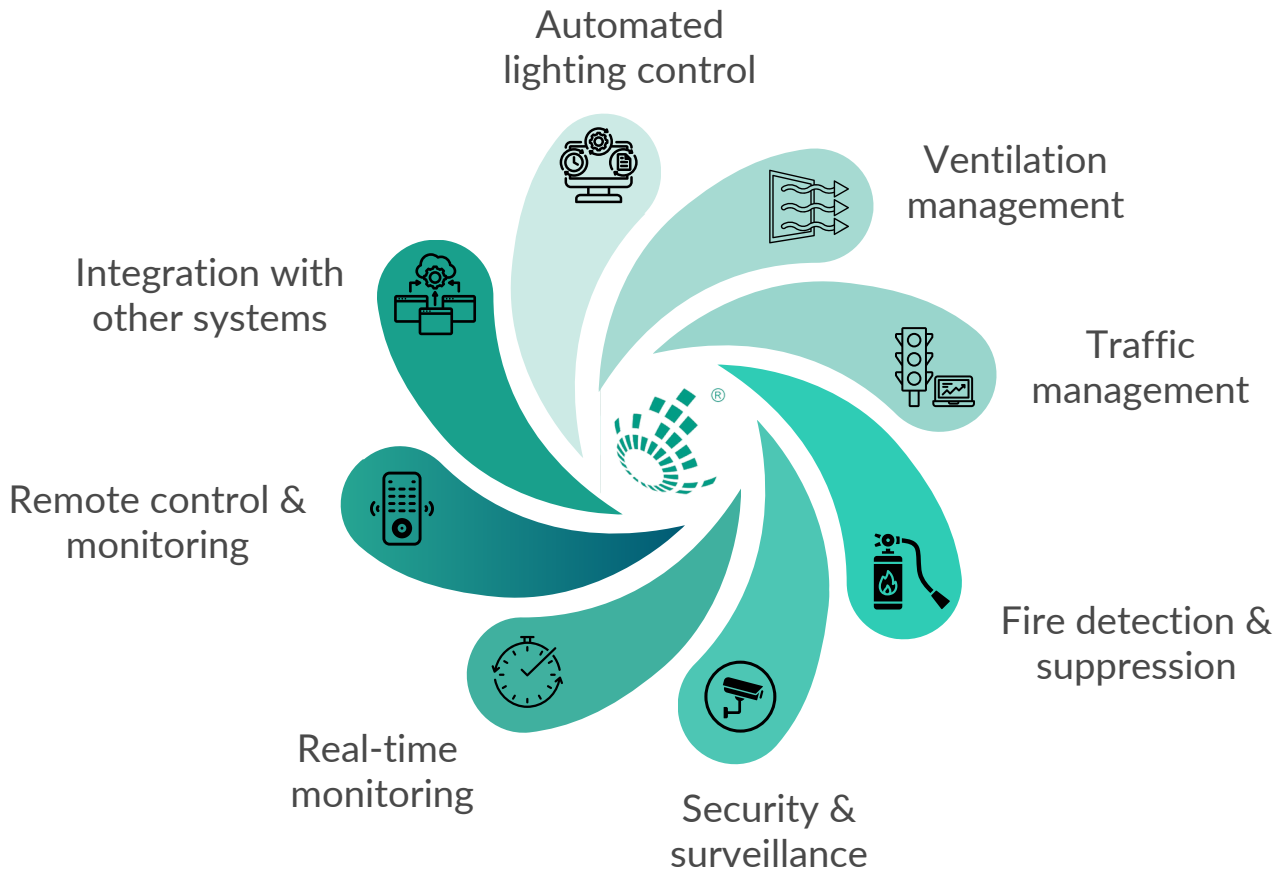
Emergency Telephones

- Emergency calling
- Location identification
- Quick response
- Assistance
- Incident reporting
- Evacuation instruction
- Medical emergencies

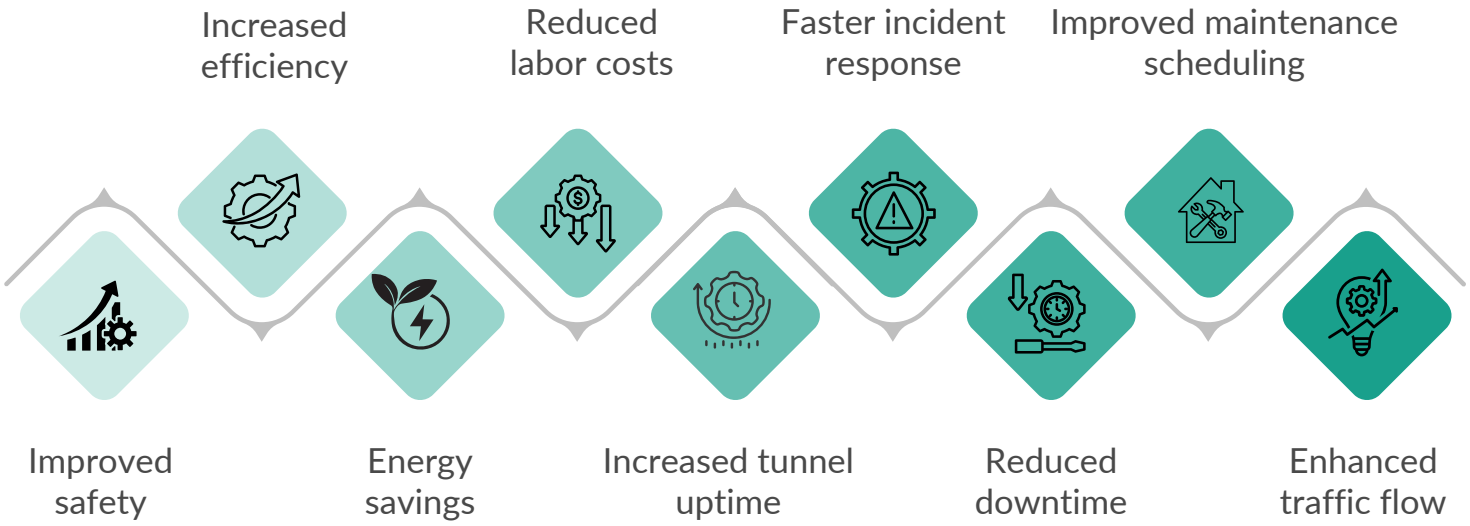


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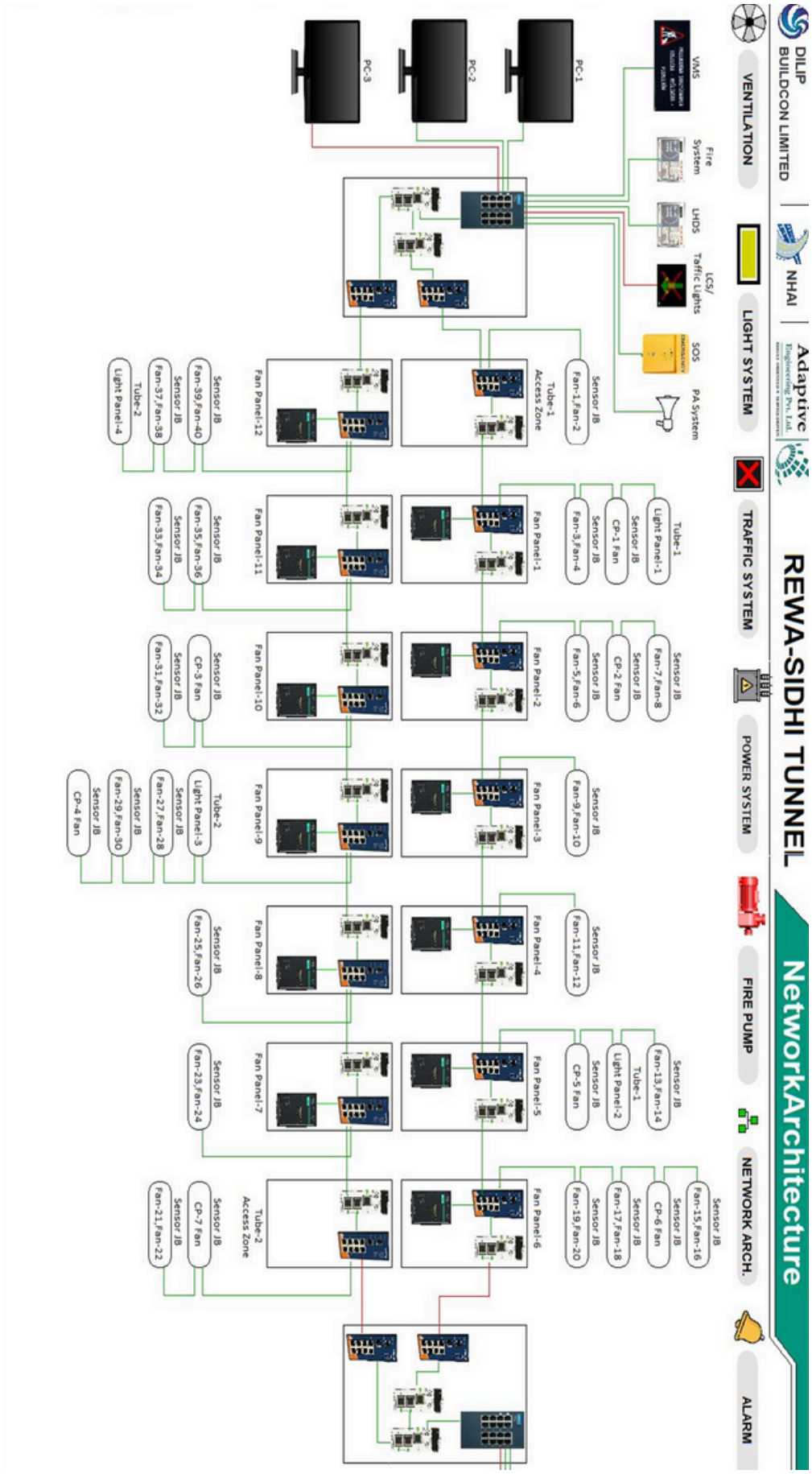
Features



Benefits



Architecture



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Value Creation



Scalable SCADA architecture

Cloud competency – secure and exclusive data

Anomaly detection to get real time fault alerts through SMS & email to concern person

Reliable communication network with low latency

Data analytics – facilitating decision making

Real time monitoring

Reduce energy consumption

Increase productivity

Better control of air quality standards

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